THIS IS

EthnoSynergy AI

Ancient formulas. Modern molecules.

AI that reads traditional polyherbal recipes and surfaces cross-plant molecular findings, each one traced to measured evidence.

THE HISTORICAL RECORD

Traditional medicine works in combinations.

Ayurvedic, Unani and TCM formulas mix several plants on purpose. Centuries of knowledge sit in manuscripts.

THE MOLECULAR RECORD

Modern research studies one plant, one compound at a time.

Like judging a band by listening to each member alone, in a separate room.

AND THE DATA IS SPLIT, TOO

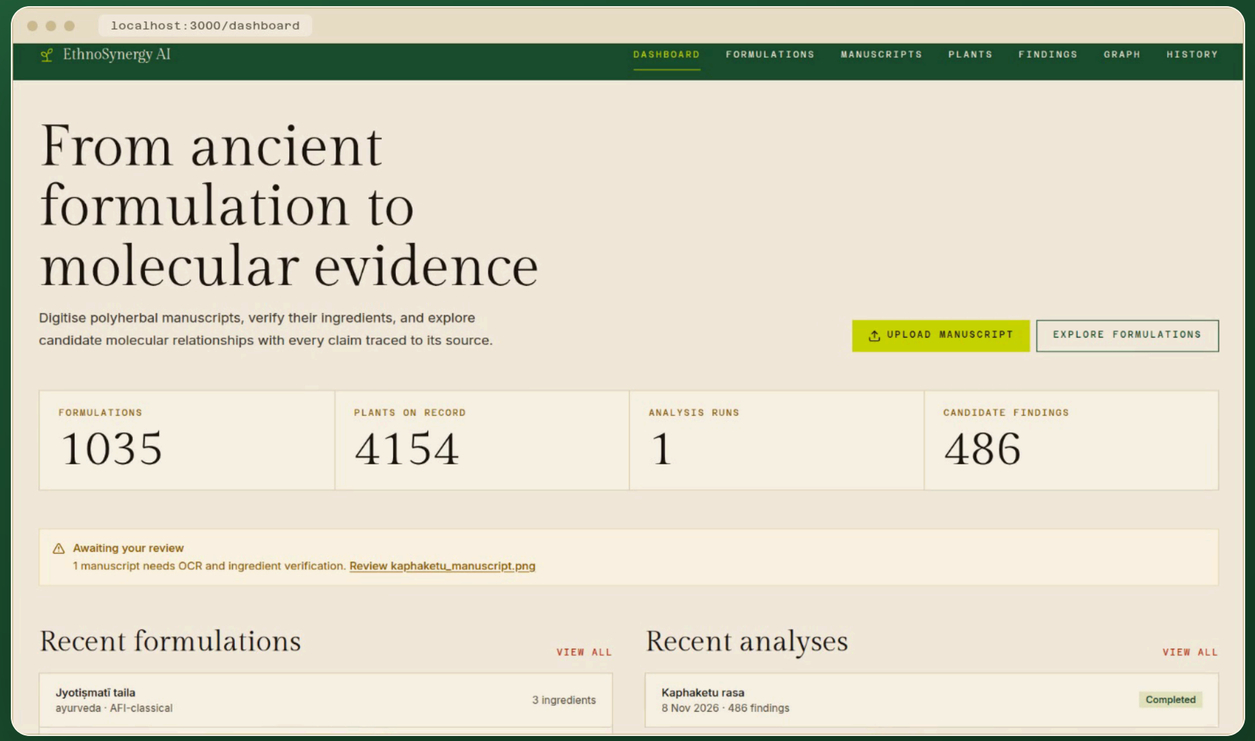
Recipes live in **manuscripts**. Compounds live in **scientific databases**. Connecting them by hand takes days.

OUR SOLUTION

So we built one place for it.

Give it a polyherbal formulation. Get back compound pairs from *different* plants that hit the *same* biological target, ranked, with measured potency and testable hypotheses.

ANCIENT RECIPE → MOLECULAR GRAPH →
CROSS-PLANT FINDINGS → HYPOTHESES



METHODOLOGY

From manuscript to mechanism in five steps.

01 · READ

OCR the recipe

Gemini reads scanned manuscripts and extracts each ingredient.

02 · IDENTIFY

Name the plants

Traditional names map to botanical ones with confidence. Low confidence is flagged uncertain for a human.

03 · GRAPH

Plants → Compounds → Targets

Pull each plant's chemistry and the proteins those molecules act on.

04 · DETECT

Find shared targets

Pairs from different plants hitting one target, ranked by `score × √shared`, checked against permutation controls.

05 · INTERPRET

Write the hypothesis

Qwen on Groq proposes a mechanism, three wet-lab experiments and the limitations.

STACK Next.js 16 · React 19 · FastAPI · PostgreSQL 16 · Docker Compose · Gemini Flash · Qwen on Groq

THE EVIDENCE MODEL

Every link gets a label.

• Measured

LAB ASSAY VALUE · IC50, EC50, KI

• Curated

EXPERT-CURATED DATABASE LINK

• Inferred

DERIVED BY RULE FROM OTHER DATA

• Predicted

MODEL OUTPUT, NEVER SHOWN AS FACT

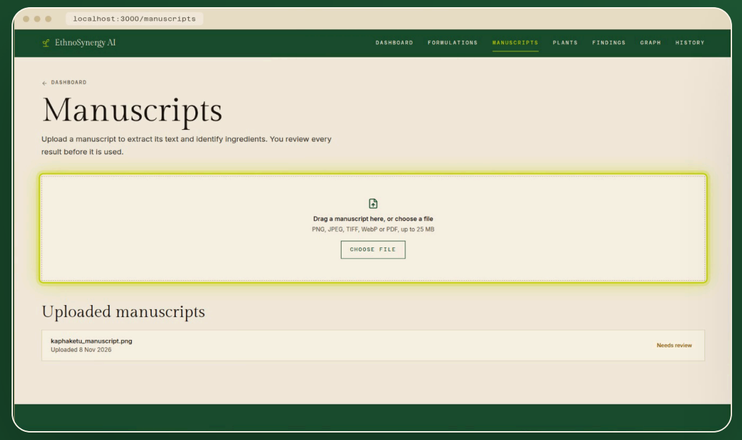
A finding is only as strong as its weakest link. Every claim traces back to a database row and its source, so prediction is never confused with measurement.

OUR DATA COMES FROM

IMPPAT 3.0	INDIAN MEDICINAL PLANTS
ChEMBL	BIOACTIVITY
NPASS	NATURAL PRODUCT ACTIVITY

4,154	PLANTS
18,304	COMPOUNDS
196,181	EVIDENCE EDGES
1,035	FORMULATIONS

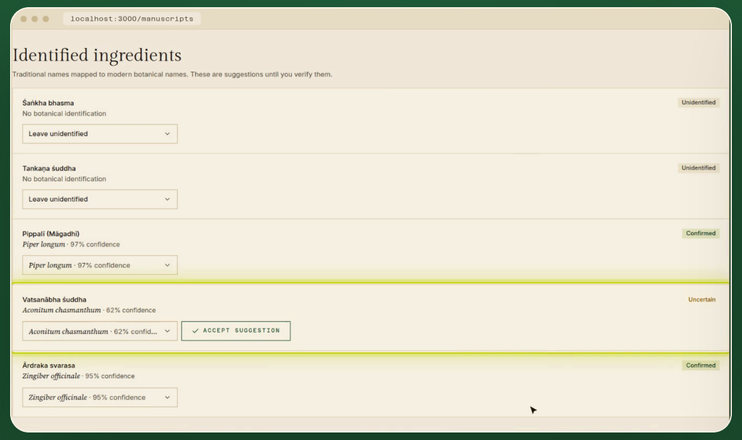
DEMONSTRATION · *Kaphaketu rasa*, A CLASSICAL AYURVEDIC RECIPE



STEP 1

Upload the manuscript

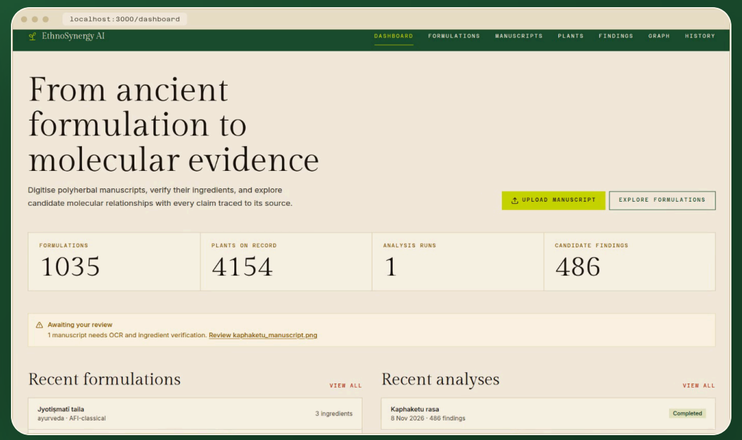
A researcher drops in a scan. Gemini reads it and lists the ingredients.



STEP 2

Flagged uncertain → I confirm it

Pippali → *Piper longum* 97%. Vatsanābha → *Aconitum* 62%, held for review.



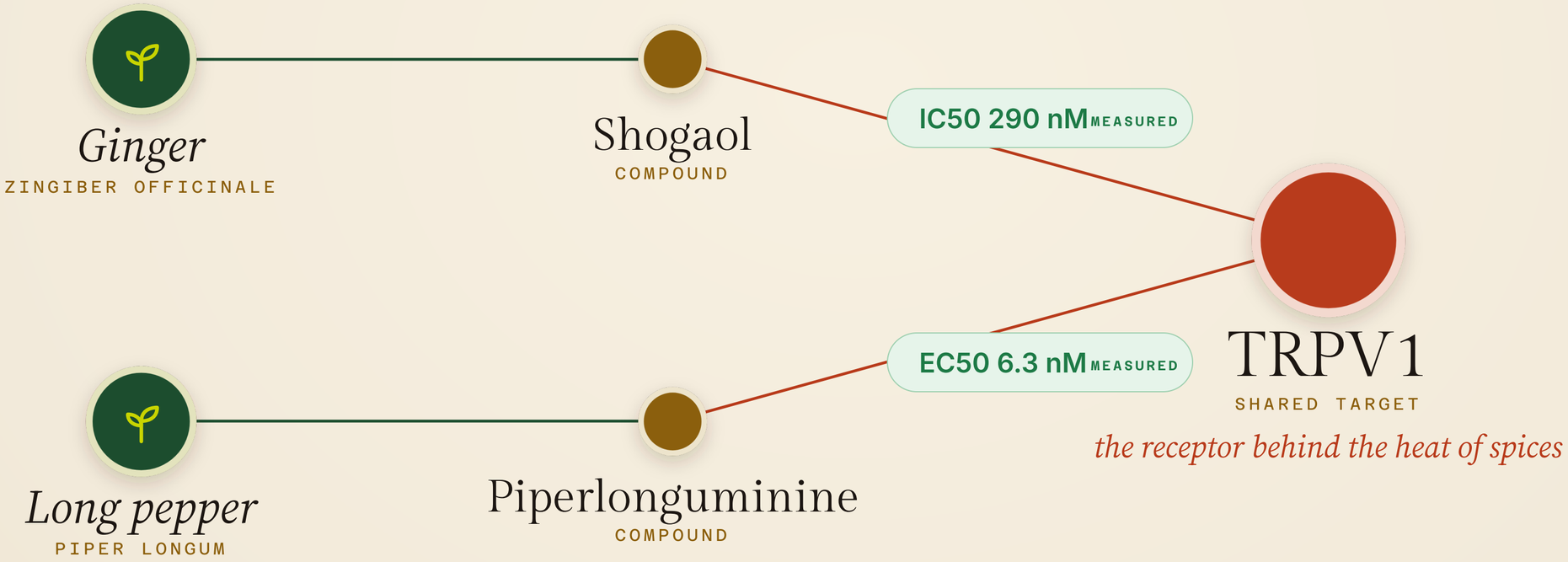
STEP 3

Run the analysis

Findings, graph, plants and history in one dashboard, each claim linked to its source.

FULL WALKTHROUGH: ETHNOSYNERGY_AI_DEMO.MP4

RESULT · NOW THE GRAPH



Two plants. Two molecules. One shared target.

KAPHAKETU RASA · IN TOTAL

3
PLANTS

403
COMPOUNDS

485
CANDIDATE RELATIONSHIPS

79
BACKED BY MEASURED LAB VALUES

MOLECULAR DATA COVERAGE

66.7%

Ginger	✓ TARGETS FOUND
Long pepper	✓ TARGETS FOUND
Vatsanābha	⚠ NO TARGET DATA

Gaps are reported, never hidden.

LLM INTERPRETATION · THE TRPV1 FINDING

“Both may act as ligands for the vanilloid receptor, with Piperlonguminine potentially showing higher affinity.”

01 Competitive binding assay for binding-site overlap

02 Calcium flux assay in TRPV1-transfected cells

03 Cryo-EM structural modelling of both binding poses

LIMITATION: IN VITRO BINDING DOES NOT CONFIRM IN VIVO ENGAGEMENT

Candidates for investigation.

Not treatments. Every output is a research proposal, never a medical claim.

> *Vatsanābha is aconite.*
> Highly toxic.
>
> That is why uncertain identifications wait for a human, and why we never suggest anyone take these remedies.

POTENTIAL IMPACT

FOR RESEARCHERS

Days of cross-referencing in seconds

Ranked, sourced candidates instead of manual database searches.

FOR LABS

Better wet-lab priorities

Each finding ships with concrete experiments and known limits.

NEXT

More formulations

All 1,035 recipes, then TCM, Unani and regional pharmacopoeias.

The point isn’t more data.

Old knowledge deserves careful evidence.